Half Adder and Full Adder Circuits

1. Half Adder Circuit

A half adder is a combinational logic circuit used to add two single-bit binary numbers. It has two inputs, A and B, and two outputs, Sum (S) and Carry (C).

Boolean Expressions:

Sum (S) = $A \oplus B$

Carry (C) = $A \cdot B$

Logic Gates Used: One XOR gate and one AND gate.

Truth Table:

Input A	Input B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Working: When both inputs are 1, the sum is 0 and the carry is 1. A half adder cannot accept a carry input from a previous stage.

Applications:

1. Binary addition.
2. Arithmetic circuits.
3. Digital calculators.

2. Full Adder Circuit

A full adder is a combinational logic circuit used to add three single-bit binary inputs: A, B, and Carry-in (Cin). It produces two outputs: Sum (S) and Carry-out (Cout).

Boolean Expressions:

Sum (S) = $A \oplus B \oplus Cin$

Carry-out (Cout) = $AB + ACin + BCin$

Logic Gates Used: Two XOR gates, two AND gates, and one OR gate.

Truth Table:

Input A	Input B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Working: A full adder adds three input bits and generates a sum and carry. It can be constructed using two half adders and one OR gate.

Applications:

1. Multi-bit binary addition.
2. Arithmetic Logic Units (ALUs).
3. Computers and digital processors.